

Polyregular Classification of Additive-Level Sorts with Coprime Weights of Absolute Value at Most Two

Xiaojian Zeng

PBC School of Finance, Tsinghua University, 100190

xiaojianzeng@foxmail.com

Abstract

An additive-level sort assigns an integer weight to each up- and down-step of a Dyck word and stably sorts the steps by their starting levels, using either left-to-right or right-to-left order to break ties. We classify all such sorts for coprime integer weights (u, d) satisfying $|u|, |d| \leq 2$. For either tie order, the restriction to Dyck words admits a partial polyregular realization exactly when $ud \geq 0$. The positive cases are realized by identity, reversal, or an explicit one-copy MSO transduction that reverses constant-level plateaux. Every mixed-sign case is excluded by a regular-slice obstruction: three elementary Dyck families turn fixed regular output probes into nonregular comparisons between two unary parameters. The classification covers all 32 labeled weight–tie-order cases; the broader classification of unrestricted additive weights is not addressed.

1 Definition and classification

Let $\Sigma = \{U, D\}$. For a word $w = w_1 \cdots w_N$, put

$$h_w(k) = \#_U(w_1 \cdots w_k) - \#_D(w_1 \cdots w_k).$$

The Dyck language is

$$\mathcal{D} = \{w \in \Sigma^* : h_w(k) \geq 0 \text{ for all } k, h_w(|w|) = 0\}.$$

Fix integer weights (u, d) and write $\nu(U) = u$, $\nu(D) = d$. The starting level of position i is

$$\ell_{u,d}^w(i) = \sum_{j < i} \nu(w_j).$$

For the tie order L , sort positions by $(\ell(i), i)$; for R , sort them by $(\ell(i), -i)$. Reading the letters in the resulting order defines $\Phi_{u,d}^L(w)$ and $\Phi_{u,d}^R(w)$. This is the additive-level sort of Baek, Hwang, La, and Yang [1], with the scan direction made explicit.

The admissible weight pairs in this note are

$$W = \{(1, 1), (1, -1), (-1, 1), (-1, -1), (1, 2), (1, -2), (-1, 2), (-1, -2), \\ (2, 1), (2, -1), (-2, 1), (-2, -1), (1, 0), (-1, 0), (0, 1), (0, -1)\}.$$

Thus (u, d) is coprime, $|u|, |d| \leq 2$, and $(u, d) \neq (0, 0)$. We use the realization convention from [1]: a positive result means that some partial polyregular map on Σ^* is defined on every word in $\mathcal{D}$ and agrees there with the level sort. Polyregular functions and their equivalent machine/logical presentations are developed in [2, 3].

Theorem 1 (Complete bounded classification). *For every $(u, d) \in W$ and each $t \in \{L, R\}$,*

$$\Phi_{u,d}^t \upharpoonright_{\mathcal{D}} \text{ has a partial polyregular realization } \iff ud \geq 0.$$

Here is the full labeled table. The symbols PR and NPR mean, respectively, polyregular-realizable and not polyregular-realizable on the Dyck domain.

(u, d)	L	R	certificate
$(1, 1)$	PR	PR	id
$(1, -1)$	NPR	NPR	$P_{m,n}$ parity probes
$(-1, 1)$	NPR	NPR	negation symmetry from $(1, -1)$
$(-1, -1)$	PR	PR	rev
$(1, 2)$	PR	PR	id
$(1, -2)$	NPR	NPR	$Z_{m,n}$ suffix probes
$(-1, 2)$	NPR	NPR	negation symmetry from $(1, -2)$
$(-1, -2)$	PR	PR	rev
$(2, 1)$	PR	PR	id
$(2, -1)$	NPR	NPR	$Y_{m,n}$ suffix probes
$(-2, 1)$	NPR	NPR	negation symmetry from $(2, -1)$
$(-2, -1)$	PR	PR	rev
$(1, 0)$	PR	PR	id, M_U
$(-1, 0)$	PR	PR	rev $\circ M_U$, rev
$(0, 1)$	PR	PR	id, M_D
$(0, -1)$	PR	PR	rev $\circ M_D$, rev

In each of the last four rows, the two listed certificates are in L, R order. The remainder of the paper proves the theorem and makes every certificate explicit.

2 The polyregular cases

We first record a symmetry valid on every input word. Write $\bar{L} = R$ and $\bar{R} = L$.

Lemma 2 (Simultaneous negation). *For every pair of integer weights and either tie order,*

$$\Phi_{-u,-d}^t = \text{rev} \circ \Phi_{u,d}^{\bar{t}}.$$

Proof. Negating all levels reverses the order of the level blocks. Reversing the entire output reverses both the order of those blocks and the order within each block. The latter reversal changes L ties into R ties and conversely. $\square$

It remains to give the plateau maps in the zero-weight cases. In the usual word structure, with position order $<$ and unary letter predicates U, D , fix $A \in \{U, D\}$ and define, for $x < y$,

$$I_A(x, y) \iff \exists z (x \leq z < y \wedge A(z)).$$

Define a binary relation on positions by

$$x \prec_A y \iff (x < y \wedge I_A(x, y)) \vee (y < x \wedge \neg I_A(y, x)). \quad (1)$$

Let M_A be the one-copy MSO string transduction that selects every input position, preserves its label, and uses $\prec_A$ as its output order.

Lemma 3 (Plateau reversal). *The relation $\prec_A$ is a strict total order. The transduction M_A lists, from left to right, the intervals on which the number of preceding A 's is constant and reverses the positions inside each such interval. Consequently M_A is a regular, hence polyregular, string function.*

Proof. Two positions lie in distinct such intervals exactly when the earlier position and the gap following it contain an A . The first disjunct of (1) then orders their intervals from left to right. If there is no such A , the second disjunct reverses their input order inside one interval. This description immediately gives irreflexivity, totality, and transitivity. It is a first-order description of a one-copy MSO transduction. $\square$

If $u, d > 0$, the starting level strictly increases at every position, so both tie orders give the identity. If $(u, d) = (1, 0)$, the level is the number of preceding U 's; if $(u, d) = (0, 1)$, it is the number of preceding D 's. Therefore, on all input words,

$$\begin{array}{c|cc} (u, d) & L & R \\ \hline (1, 0) & \text{id} & M_U \\ (0, 1) & \text{id} & M_D. \end{array} \quad (2)$$

Identity, reversal, and the maps M_A are regular. Regular functions are closed under composition. Lemma 2 and (2) now give the displayed certificate for every one of the 20 labeled cases with $ud \geq 0$.

3 A regular-slice obstruction

All lower bounds use the following criterion. It is the regular-slice form of the linear-growth argument used in Proposition 5.6 of Baek–Hwang–La–Yang [1].

Proposition 4 (Regular-slice criterion). *Suppose a deterministic two-way finite-state transducer maps $a^m \# b^n$, for $m, n \geq 1$, to a Dyck word $W_{m,n}$ of length $O(m + n)$. If some regular language $K \subseteq \Sigma^*$ has a nonregular pullback*

$$\{a^m \# b^n : m, n \geq 1, \Phi_{u,d}^t(W_{m,n}) \in K\},$$

then $\Phi_{u,d}^t \upharpoonright_{\mathcal{D}}$ has no partial polyregular realization.

Proof. Precompose a hypothetical realization with the two-way transducer and restrict to the regular parameter language. The resulting polyregular function is total there and has linear output growth because every level sort is length-preserving and $|W_{m,n}| = O(m + n)$. The linear-growth collapse for polyregular functions makes this a regular/MSO string function. Inverse images of regular languages under regular string functions are regular, a contradiction. $\square$

We will use the two nonregular unary-comparison languages

$$L_{\geq c} = \{a^m \# b^n : m, n \geq 1, n \geq m + c\} \quad (c \geq 0), \quad (3)$$

$$L_{\leq} = \{a^m \# b^n : m, n \geq 1, n \leq m\}. \quad (4)$$

For completeness, if a pumping length is p , pumping up inside the first block of $a^p \# b^{p+c}$ disproves regularity of $L_{\geq c}$, while pumping down inside the first block of $a^p \# b^p$ disproves regularity of $L_{\leq}$. Imposing fixed positive lower bounds on m, n does not change either proof.

4 The height weights $(1, -1)$

Let

$$P_{m,n} = U^m D^m U^n D^n \in \mathcal{D}.$$

A deterministic two-way transducer generates this word from $a^m \# b^n$ by making an up-step pass and a down-step pass over each unary block. Directly grouping the four runs by their starting heights gives

$$\Phi_{1,-1}^R(P_{m,n}) = \begin{cases} UU(DU)^{2n-2}DD(UD)^{m-n}, & n \leq m, \\ UU(DU)^{2m-1}DD(UD)^{n-m-1}, & n > m, \end{cases} \quad (5)$$

and

$$\Phi_{1,-1}^L(P_{m,n}) = \begin{cases} UU(UD)^{2m-2}D(UD)^{n-m}D, & m \leq n, \\ UU(UD)^{2n-2}UDD(UD)^{m-n-1}D, & m > n. \end{cases} \quad (6)$$

Indeed, level zero supplies the initial UU . At every positive level in which all four runs occur, decreasing input position supplies $DUDU$ and increasing input position supplies $UDUD$. Once the shorter pyramid ends, the remaining two-run blocks give the central DD and the displayed tail. This accounts for every level and proves both formulas.

For right ties, take the regular language

$$K_R = \{UU(DU)^{2q}DD(UD)^s : q, s \geq 0\}.$$

The displayed DD is the first DD after the initial UU , so the parse is unique. Equation (5) gives

$$\Phi_{1,-1}^R(P_{m,n}) \in K_R \iff n \leq m.$$

Thus its pullback is (4).

For left ties, restrict harmlessly to $m, n \geq 2$ and use

$$K_L = \{UU(UD)^{2q+1}UDD(UD)^s D : q, s \geq 0\}.$$

In the first branch of (6), rewrite $(UD)^{2m-2}D = (UD)^{2m-3}UDD$; the exponent preceding the first UDD is odd. In the second branch it is $2n-2$, which is even. Again the first DD fixes the parse, and hence

$$\Phi_{1,-1}^L(P_{m,n}) \in K_L \iff m \leq n.$$

The pullback is $L_{\geq 0}$. Proposition 4 proves non-polyregularity for both tie orders at $(1, -1)$.

5 The asymmetric mixed-sign weights

The remaining representatives are separated by fixed output suffixes. We include the complete starting-level tables and the boundary analysis, since the ties at the top few levels determine the thresholds.

5.1 Weight $(1, -2)$

Put

$$Z_{m,n} = (UD)^m U^n D^n \in \mathcal{D}, \quad r = n - m.$$

This family is generated from $a^m \# b^n$ by outputting UD per a and making one U -pass and one D -pass over the b -block. Its four kinds of steps have starting levels

step	index range	starting level
U in $(UD)^m$	$1 \leq j \leq m$	$1 - j$
D in $(UD)^m$	$1 \leq j \leq m$	$2 - j$
U in U^n	$1 \leq k \leq n$	$k - m - 1$
D in D^n	$1 \leq k \leq n$	$r - 2k + 2$.

(7)

It follows that

$$\Phi_{1,-2}^L(Z_{m,n}) \in \Sigma^*UD \iff r \geq 2, \quad (8)$$

$$\Phi_{1,-2}^R(Z_{m,n}) \in \Sigma^*UUD \iff r \geq 4. \quad (9)$$

Here are the terminal-block details. If $r \geq 2$, the unique top step is the first mountain D , at level r , and the preceding block ends in the last mountain U ; this remains true at $r = 2$, where an old zigzag D ties below it. If $r \leq 1$, the top block is at level one and ends in a zigzag D , while the preceding letter is also D . This proves (8).

For right ties and $r \geq 4$, the top three relevant blocks, beginning at level $r - 2$, end as DU, U, D , and therefore the word ends in UUD . At $r = 3$ they instead give DUD ; at $r = 2$ and $r = 1$ the word ends in DD . For $r \leq 0$, the right-tie level-zero block ends in DU immediately before the top zigzag D , so the suffix is not UUD . This proves (9).

The pullbacks in (8)–(9) are respectively $L_{\geq 2}$ and $L_{\geq 4}$. Proposition 4 settles both tie orders for $(1, -2)$.

5.2 Weight $(2, -1)$

Put

$$Y_{m,n} = U^m D^m (UD)^n \in \mathcal{D}, \quad r = n - m.$$

A two-way transducer makes an up-pass and a down-pass over the a -block and then outputs UD per b . The starting levels are

step	index range	starting level
U in U^m	$1 \leq i \leq m$	$2(i - 1)$
D in D^m	$1 \leq i \leq m$	$2m - i + 1$
U in $(UD)^n$	$1 \leq j \leq n$	$m + j - 1$
D in $(UD)^n$	$1 \leq j \leq n$	$m + j + 1$.

(10)

Grouping the largest values in (10) yields

$$\Phi_{2,-1}^L(Y_{m,n}) \in \Sigma^*DUDD \iff r \geq 1, \quad (11)$$

$$\Phi_{2,-1}^R(Y_{m,n}) \in \Sigma^*DUDUDDD \iff r \geq 3. \quad (12)$$

For (11), if $r \geq 1$, the top two levels are singleton zigzag D 's. The preceding level ends in a DU block; when $r = 1$, the top of the mountain supplies the needed D at the boundary. Thus the suffix is $DUDD$. If $r = 0$, the block immediately below the top consists only of D 's; for $m = n \geq 2$ the output has at least three terminal D 's, while $m = n = 1$ gives UDD . If $r \leq -1$, the maximum mountain levels again force at least three terminal D 's. None of these words ends in $DUDD$.

For (12), when $r \geq 3$ the five relevant top blocks, read from lower to higher level, end as

$$D, \quad UD, \quad UD, \quad D, \quad D,$$

which gives $DUDUDDD$. All remaining boundary regimes are distinguished directly by (10):

r	terminal obstruction to $DUDUDDD$
2	the boundary block is UDD ; $(m, n) = (1, 3)$ ends in $UUDUDDD$
1	the boundary block is UDD ; $(m, n) = (1, 2)$ ends in $UUUDDD$
0	at least four terminal D 's; $(1, 1)$ gives $UUDD$
-1	the top tie and the next level are D -blocks; $(2, 1)$ gives $UUUDDD$
-2	the top blocks end DDU, DD, D ; $(3, 1)$ ends in $UUDUDDD$
-3	the decisive top blocks end DDU, D, D
≤ -4	the decisive top blocks end DU, D, D .

The listed block endings are obtained simply by solving equality with each of the four arithmetic progressions in (10); they exhaust all $r \leq 2$ and are incompatible with the target suffix. This proves (12).

The two pullbacks are $L_{\geq 1}$ and $L_{\geq 3}$, so Proposition 4 proves non-polyregularity for both tie orders at $(2, -1)$.

6 Completion, scope, and verification

The preceding sections give lower bounds for the six representatives $(1, -1, t)$, $(1, -2, t)$, and $(2, -1, t)$ with $t \in \{L, R\}$. Lemma 2 transports each lower bound to the simultaneously negated weight pair, with the opposite tie order. These are exactly the 12 labeled cases with $ud < 0$. Together with the 20 explicit realizations from Section 2, this proves Theorem 1.

The closest comparison is the framework of Baek–Hwang–La–Yang [1]. Their paper defines the additive-level sorts, observes the identity case, proves the right-tie $(1, -1)$ lower bound, and asks which additive-level sorts are polyregular. The contribution here is the complete finite sub-classification for the specified 16 small coprime weight pairs, including the other tie order, the four asymmetric representative obstructions, and explicit realizations for all zero-weight cases. No conclusion is claimed for weights outside this bounded family.

As a transcription check, an exact-integer program evaluated the two closed forms (5)–(6) for all $1 \leq m, n \leq 80$ (6,400 pairs) and the four suffix equivalences (8), (9), (11), and (12) for all $1 \leq m, n \leq 160$ (25,600 pairs). These finite checks are not used as a proof; the all-parameter arguments are the level tables and pumping obstructions above.

An AI system assisted with organizing the case analysis, drafting the manuscript, and producing the exact finite regression checks. The author reviewed the stated definitions, formulas, boundary cases, and proof scope; no numerical or model-generated confidence assessment is treated as mathematical evidence.

References

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